

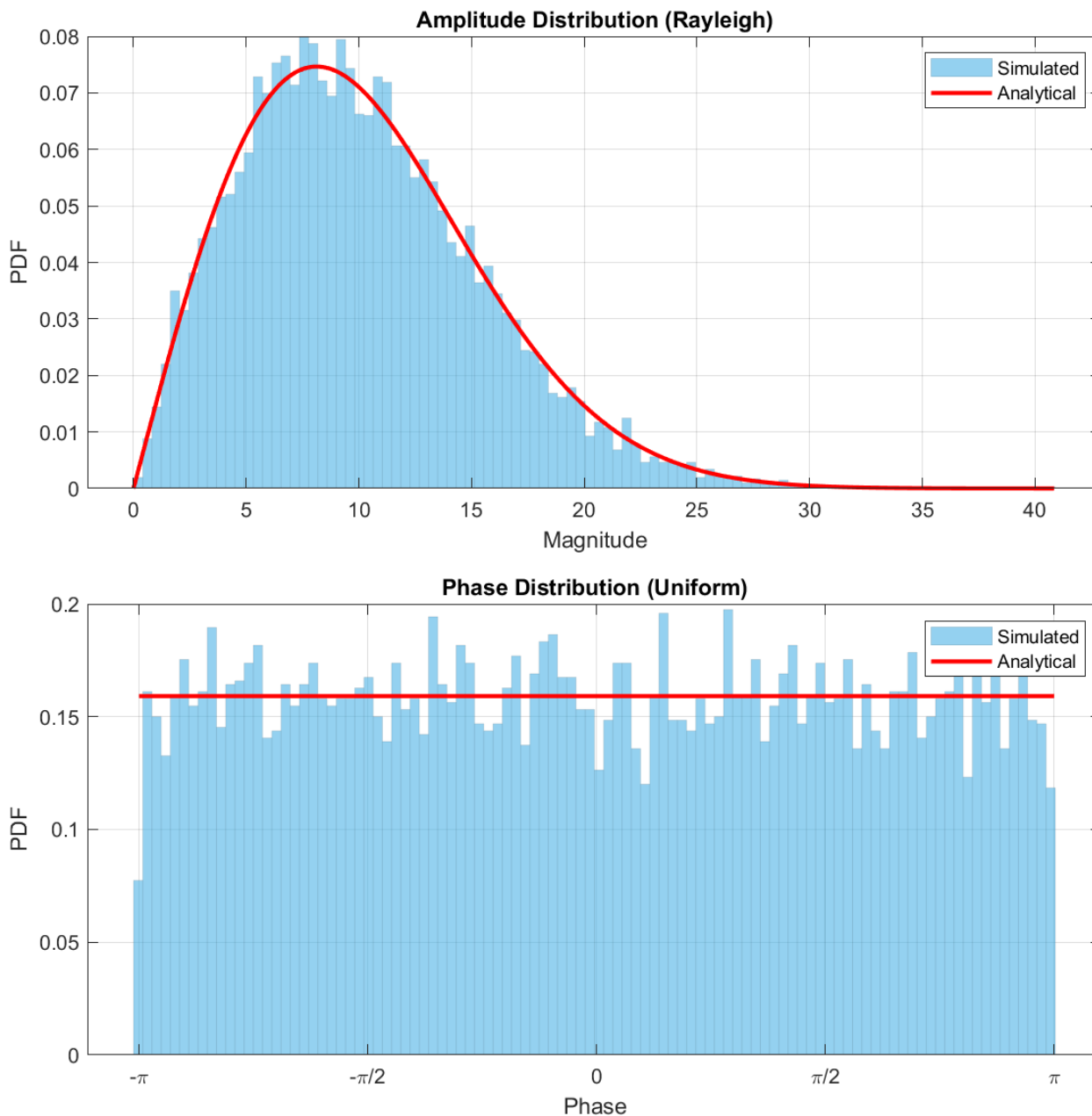
EE 632

Assignment 1

MATLAB Part

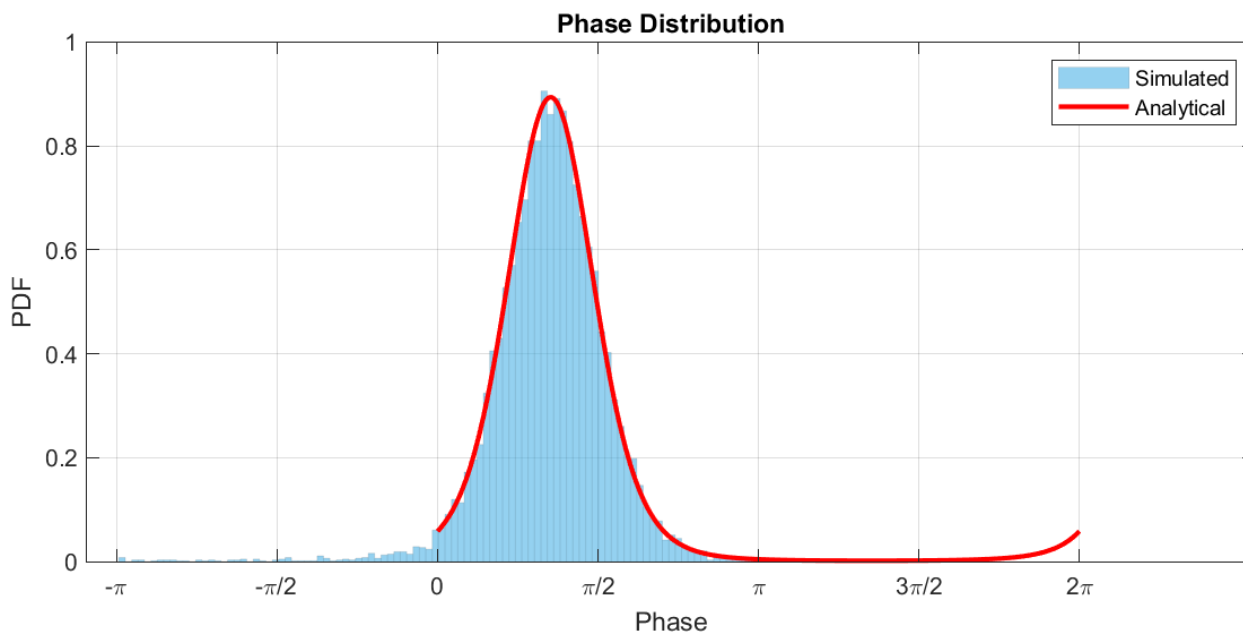
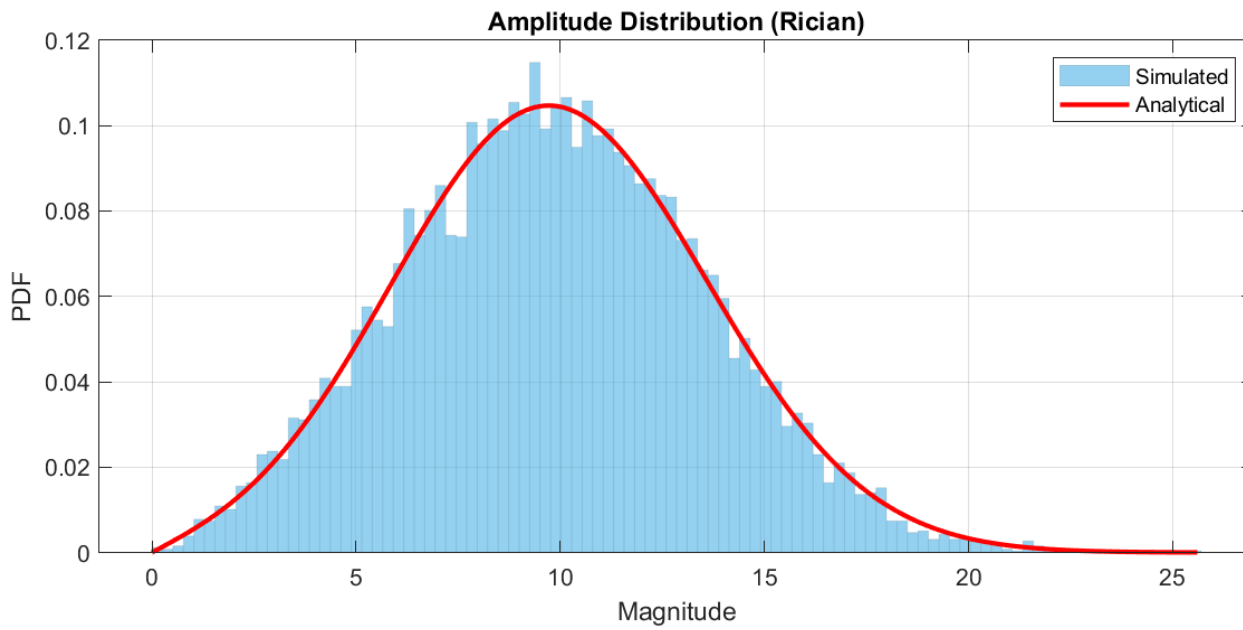
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Q3



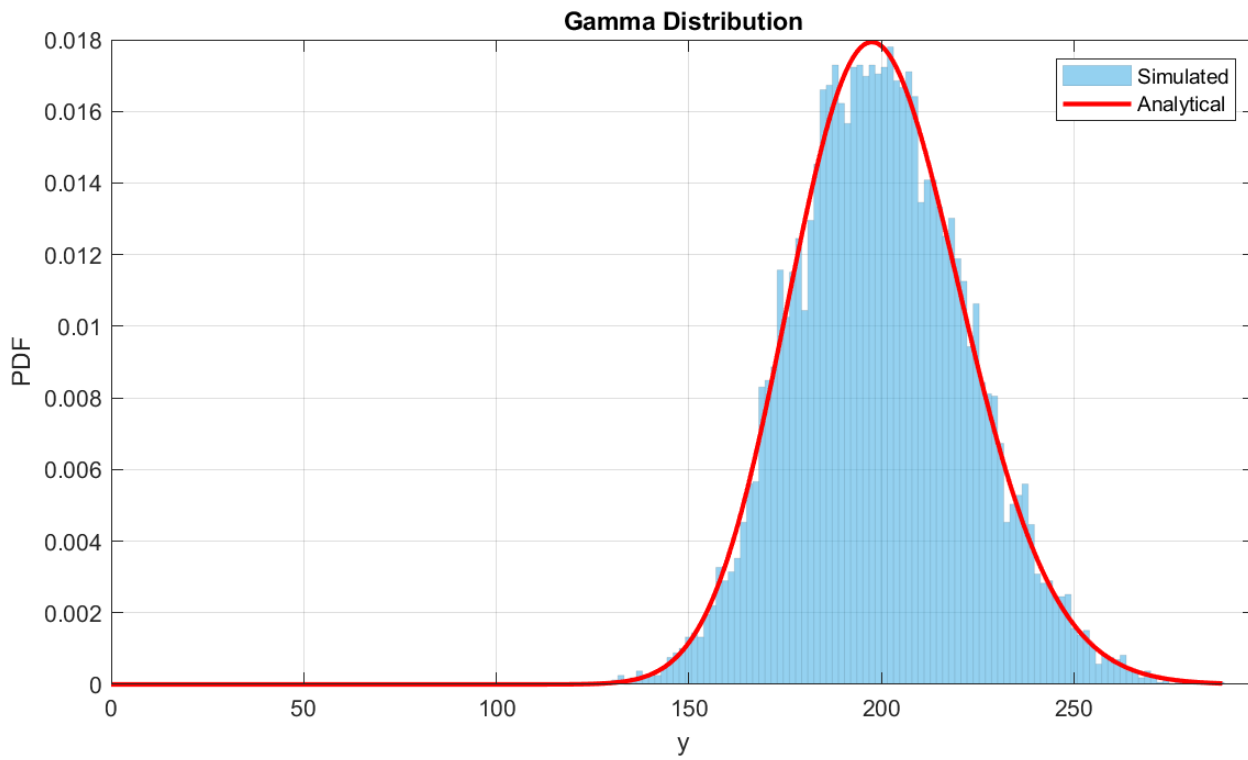
From the plots we infer that amplitude of RV X follows Rayleigh distribution while the phase follows uniform distribution.

Q4



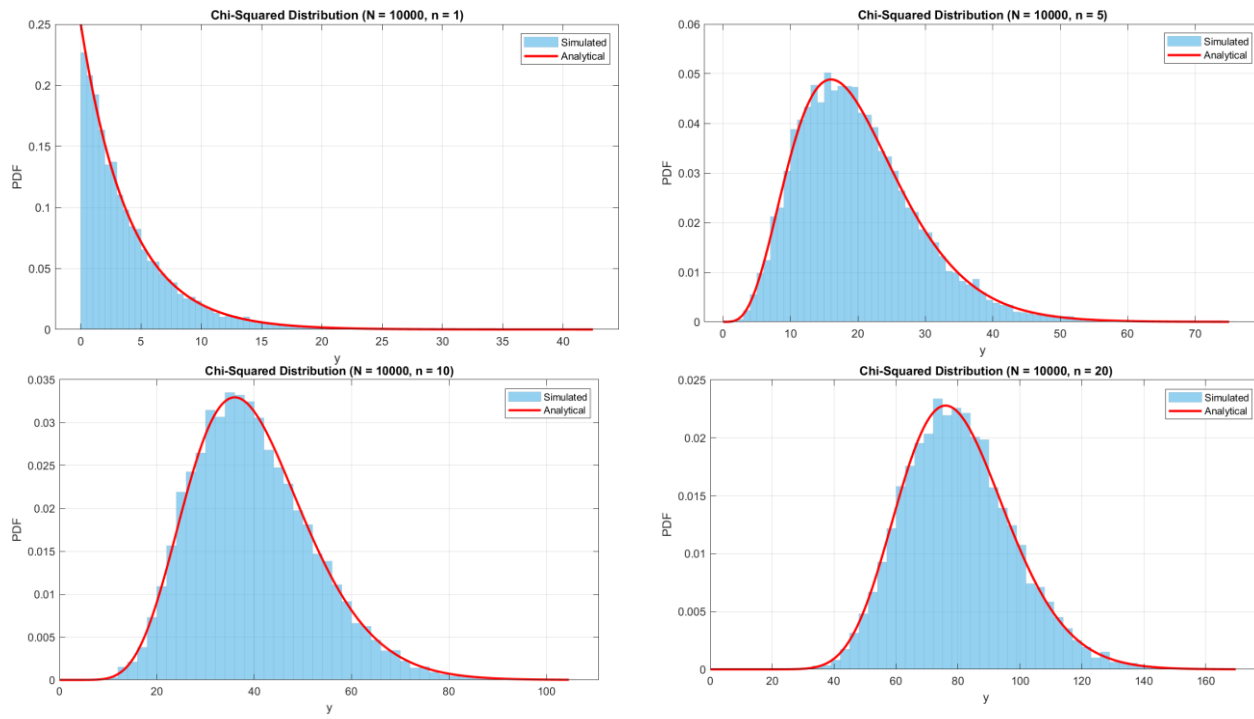
From the plots we infer that amplitude of RV X follows Rician distribution while the phase follows a very complex distribution as given in the question.

Q5

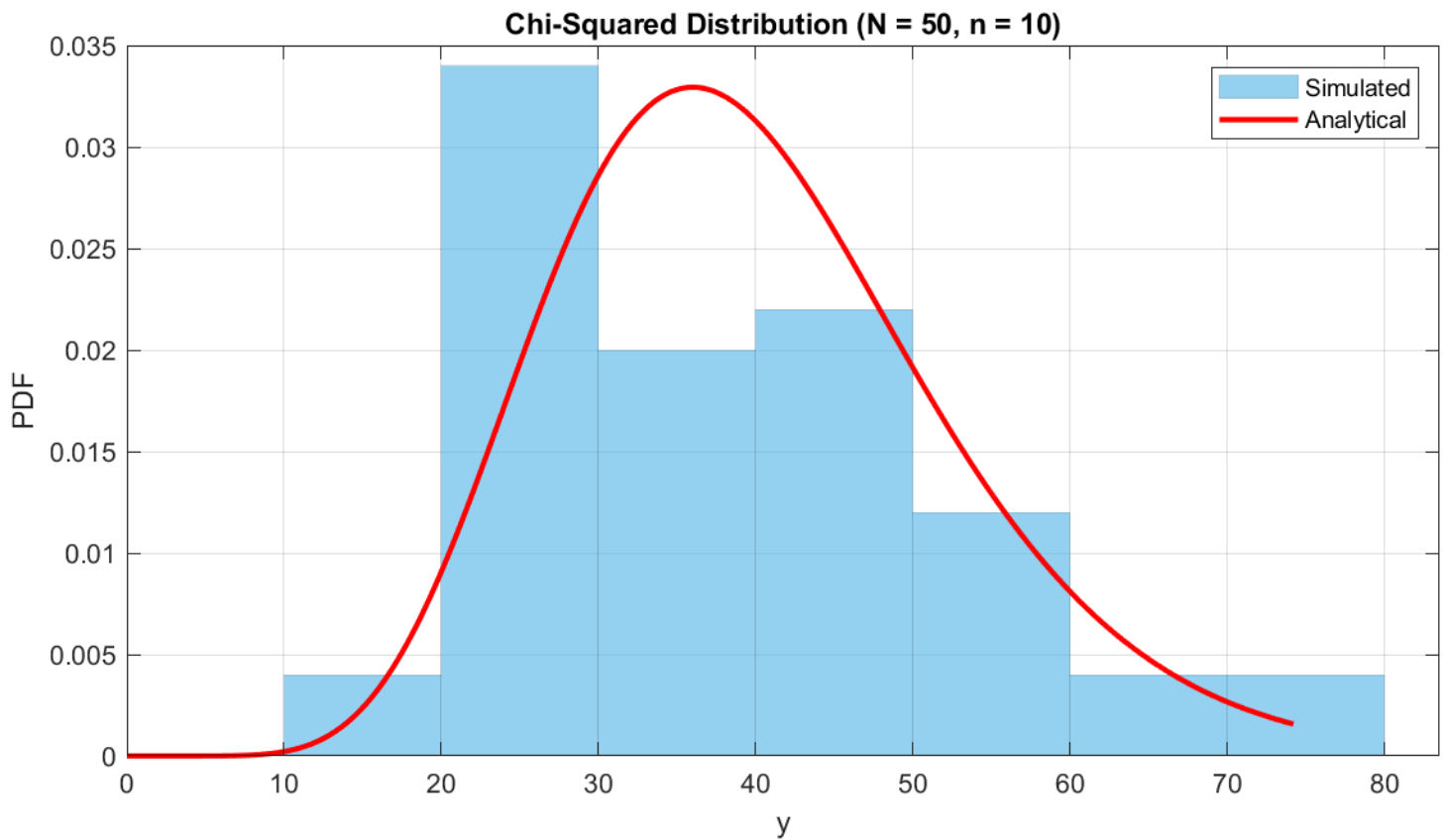


We can observe that the simulated PDF plot follows the exponential distribution.

Q6



By changing n we change the degrees of freedom of RV. An increase in n leads to rightward shift in the plot as we observe the peak lowers, for high n the distribution looks like Normal distribution (Central Limit Theorem).



By changing N we observe that below 50, the distribution fails to follow the Chi-Squared distribution.